

Reset the Entryway SOP (1x Episode: coats, shoes, keys, bag, mat, in situ)

One episode resets the whole entryway at the bench where these things live. The two coats go first. Then the two shoes. Then the keys. Then the bag. The mat is squared last. The episode ends when the bench top is clear, every thing hangs or sits where it belongs, and the mat runs straight in front of the bench. Never split the coats, the shoes, the keys, the bag, and the mat into two episodes.

The base is passive. It has no motor. You push it by hand to the bench in front of the hook board and lock it there. Nothing is carried away to another room. Everything the episode touches is already there when recording starts: the bench, the hook board above it with its three empty hooks, the two coats at the front left, the two loose shoes at the front middle, the keys at the front right, the bag past the keys, the empty bowl at the back right, the empty shoe rack on the floor at the left end, and the mat on the floor in front of the bench.

The entryway is used as found. The three hooks are empty. The bowl is empty. The shoe rack is empty. The bench top is clear but for the two coats, the two shoes, the keys, and the bag. The mat lies on the floor crooked. The episode ends with two coats on the two coat hooks, two shoes paired on the rack, the keys in the bowl, the bag on the bag hook, the bench top clear, and the mat square.

This is an in situ task. Four rules come from that. First, everything happens at this bench. Nothing is carried away and nothing new is brought in. Second, every carry goes along the front lane, low and level. You lift a thing just off what it lies on, bring it forward into the lane, and carry it along the lane close over the bench. The only lifts that go up are the coat loops and the bag strap going onto their hooks. The only reaches that go down are the shoes going onto the rack and the mat. Third, the fixtures never move. You never lean on, push, drag, or lift the bench, the hook board, a hook, the bowl, or the shoe rack. Fourth, nothing is dragged. Every pick lifts the thing straight up off the surface before it moves anywhere. The mat is the one thing that is never lifted. It is pushed flat along the floor.

Things go on the bench, on a hook, in the bowl, or on the rack, and nowhere else. Nothing is ever left on the floor but the shoes on the rack and the mat. A thing that lands on the floor is a drop. You do not pick a drop back up. You report it.

The order never changes: hang the first coat, hang the second coat, put the first shoe on the rack, pair the second shoe beside it, put the keys in the bowl, hang the bag, square the mat. Nothing starts until the thing before it is done and let go.

Only one thing is held at a time. Nothing is passed from one grip to another, and no grip is changed while a thing is off the bench. Every thing is put down or let go under control.

Setup

Do the base positioning and both checklists before you start an episode.

Cell configuration

Environment camera: 900 mm.

cell_type: bimanual

Base positioning

The base is passive. It has no motor. You push it into place by hand. You never steer it, nudge it, or move it after recording starts. You park it once, before recording, and it stays there until the episode is over.

1. Push the base by hand up to the bench. Stop it square to the front edge of the bench, so the front edge runs straight across the frame and both ends look the same distance away.
2. Stop it in the middle of the bench, so the coats at the left and the bag at the right are the same distance from the middle of the base.
3. Stop it close enough that a gripper reaches the far coat hook without stretching the arm. Stop it far enough that no arm, wrist, or part of the base touches the bench front, the hook board, or the mat while both arms work.
4. Check the height. With the base parked, a gripper lifts a coat off the bench by its loop, brings it forward into the front lane, and carries it to a hook. The wrist and the forearm must not hit the hook board.
5. Check the hooks. A gripper puts a loop over each coat hook and puts the bag strap over the bag hook, and lets go, without stretching and without hitting the hook board or the other arm.
6. Check the low reaches. A gripper reaches the shelf of the shoe rack and the near edge of the mat without stretching and without the base touching either one.
7. Check the bench. A gripper reaches both coats, both shoes, the keys, the bag, and the bowl. It does all of them without stretching and without knocking anything over.
8. Lock the base. Push it once, hard, by hand. It must not roll, slide, or turn.
9. If any line from 1 to 7 fails, push the base to a new spot by hand and start again at line 1. Do not work at a bench the arms cannot reach easily.

The base stays locked and still for the whole episode. Nothing moves it. No arm leans on the bench hard enough to push it. Nobody touches it by hand. You never move it during the task. A base that moves after recording starts ends the episode.

Hardware checklist

1. Cameras are on and recording.

2. The environment camera points at the middle of the bench. The frame shows the whole bench top, the hook board with all three hooks, the two coats, the two shoes, the keys, the bag, the bowl, the shoe rack, and the whole mat.
3. The camera sees the bench from the front, so you can see where a thing lies and whether it lies flat and square.
4. The camera sees each hook from the front, so you can see whether a loop or a strap sits all the way over it.
5. The camera sees into the bowl from the front, so you can see whether the keys are in it.
6. The camera sees the shelf of the shoe rack, so you can see how the two shoes stand and whether they are paired.
7. The camera sees the front edge of the bench and the near edge of the mat in the same frame, so you can see whether the mat runs straight.
8. Both arms are at home with the grippers open.
9. The arms reach both coats, both shoes, the keys, the bag, the bowl, all three hooks, the rack shelf, and the near edge of the mat. They reach all of them without stretching to a joint limit.
10. A gripper lifts a thing just off what it lies on and carries it low and level along the front lane. It never carries one across the back of the bench and never over the bowl.
11. The two arms do not hit each other. One arm working at a hook stays out of the other arm's path along the front lane.
12. If the arms cannot reach a place, park the base again with the Base positioning steps until lines 9 to 11 are all true.

Materials checklist

1. The bench is one long flat top set against the wall under the hook board. Nobody moves it, leans on it, or stands anything on its edge.
2. The bench top is dry, clean, and clear but for the two coats, the two shoes, the keys, and the bag.
3. The hook board is fixed to the wall above the back of the bench. It does not move, turn, or come loose. Nobody pushes it or hangs on it.
4. The hook board carries three hooks in a row: coat hook one at the left, coat hook two next to it, and the bag hook at the right end. All three are empty at the start.
5. The hooks stand out far enough that a loop or a strap sits all the way over one and stays there.
6. Each coat is a jacket with a hanging loop sewn inside the collar. The loop stands up when the coat lies on the bench.
7. The two coats lie flat at the front left of the bench, one on top of the other, each one with its loop up and clear.

8. Neither coat has anything in its pockets, and no belt, cord, or sleeve hangs off the bench edge.
9. The shoes are one pair, a left shoe and a right shoe of the same kind. They lie loose at the front middle of the bench, apart from each other and not paired.
10. Both shoes are dry and clean underneath, so they leave no mark on the bench or the rack.
11. The shoe rack is a low open rack standing on the floor at the left end of the bench, clear of it. It has one shelf. It is empty at the start. Nobody moves it, tips it, or stands on it.
12. The rack shelf is wide enough for the two shoes to stand side by side with a finger's width between them.
13. The keys are one set on one ring. They lie at the front right of the bench. No key is loose off the ring.
14. The bowl is a shallow bowl standing at the back right of the bench top. It is empty at the start. It is wide enough and deep enough that the whole key set lies inside it. Nobody moves it, lifts it, or slides it.
15. The bag is one bag with one strap. It lies at the front right of the bench, past the keys, with its strap laid out clear and not under the bag.
16. The bag is closed and holds nothing loose that can fall out when it hangs.
17. The mat is one flat mat lying on the floor in front of the bench. It starts crooked, turned off square to the bench front edge, but fully flat with no corner folded under.
18. The mat slides on the floor when it is pushed at its edge, and it does not bunch up or roll.
19. The floor around the bench is dry and clear, so nothing is in the mat's way.

Workspace layout

Nothing is marked or taped out. You judge every place below by eye, from the bench, the hook board, and the mat themselves.

Above the bench, left to right:

1. **Coat hook one**, at the left of the hook board. Empty at the start. It holds the first coat at the end. Never moved.
2. **Coat hook two**, next to coat hook one. Empty at the start. It holds the second coat at the end. Never moved.
3. **Bag hook**, at the right end of the hook board. Empty at the start. It holds the bag at the end. Never moved.

Back row of the bench top:

1. **Bowl spot**, the back right of the bench top. The bowl stands here. It is empty at the start and holds the keys at the end.

Front row of the bench top, left to right:

1. **Coat spot**, front left. The two coats lie here in one flat stack. It is empty by the end of Step 2.
2. **Shoe spot**, front middle. The two loose shoes lie here. It is empty by the end of Step 4.
3. **Key spot**, front right. The keys lie here. It is empty by the end of Step 5.
4. **Bag spot**, front right, past the keys. The bag lies here. It is empty by the end of Step 6.

On the floor:

1. **Shoe rack**, at the left end of the bench, clear of it. Empty at the start. It holds the paired shoes at the end.
2. **Mat**, in front of the bench. It starts crooked and ends square.

Front lane: the clear strip of bench top along the front edge, from the coat spot to the bag spot. Every carry goes along it. Nobody ever puts anything down in it.

No carry passes across the back of the bench or over the bowl, and nothing travels to the far side of the bench.

Vocabulary

Coat: one of the two jackets at the coat spot. Both are hung before anything else moves.

Loop: the small sewn loop inside a coat's collar. It is the only part of a coat a gripper ever closes on.

Strap: the long band on the bag. It is the only part of the bag a gripper ever closes on.

Top of the stack: the coat lying highest with nothing on it. You always take the top one. You never pull one out from under the other.

Low carry: you lift the thing just off what it lies on, bring it forward into the front lane, and carry it along the lane close over the bench and level. You never carry it across the back of the bench and never over the bowl.

Front lane: the clear strip of bench top along the front edge, from the coat spot to the bag spot. Every carry goes along it. Nobody puts anything down in it.

Hanging right: the loop or the strap sits all the way over the hook, the thing hangs straight down, and it does not swing, slide off, or lift off the hook when the gripper lets go.

Hanging clear: the hung coat or bag hangs free of the bench top, free of the wall behind it, and free of the thing hanging next to it. No sleeve, hem, or strap rests on the bench.

Standing right on the rack: the shoe stands flat on the rack shelf, upright on its own sole, fully on the shelf with no part hanging over the front edge, and it does not fall or rock when the gripper lets go.

Paired: the two shoes stand side by side on the rack shelf, both facing the same way, their heels level with each other, and a finger's width between them. Not one in front of the other. Not one on top of the other.

In the bowl: the whole key set lies inside the bowl, below its rim, with no key hanging over the edge and no key outside the bowl.

Square: the near edge of the mat runs straight across, parallel to the front edge of the bench, and both ends of that edge are the same distance from the bench.

Push flat: the closed gripper touches the mat's edge close to the floor and pushes it along the floor. It never lifts the mat, never digs under it, and never bunches it up.

Drop: a coat, a shoe, the keys, or the bag that falls on the floor. You do not pick a drop back up. You report it.

Loose thing: a thing that ends up on the bench outside its spot, or half on a hook, or beside the bowl. It is left where it lands and cleared in the reset, never on camera.

Back and clear: the arm is pulled back, so no part of it is over the bench top, the front lane, a hook, or the mat.

Steps

Step 1: Hang the first coat

Goal: the first coat hangs on coat hook one, and the coat spot holds one coat.

1.1 Lift the top coat by its loop

Close a gripper on the loop of the top coat at the coat spot.

Lift the coat straight up off the stack, just clear of the coat under it, and no higher.

Pull it forward into the front lane. Do not swing it. Do not drag it along the bench.

Check: the coat hangs straight down from its loop and the coat under it has not come with it and has not slid. If both coats came up together, lay both back on the coat spot, square them, and take the top one again by its loop.

1.2 Carry it to the hook

Carry the coat along the front lane to coat hook one in a low carry. Keep it hanging straight the whole way.

Do not carry it across the back of the bench and do not carry it over the bowl.

Bring it in front of the hook. Then lift the loop straight up, level with the hook.

Check: the coat hangs straight and still under the gripper, in front of coat hook one. No sleeve is caught on the bench, on the bag, or on the other arm. If the coat swings, hold it still and let it settle before you go on.

1.3 Put the loop on the hook

Move the loop straight back over the hook, so the hook goes through it.

Lower the loop until it sits all the way down on the hook and the coat takes its own weight.

Open the gripper and pull it back away from the coat.

Check: the coat is hanging right and hanging clear. The loop sits all the way over the hook, the coat hangs straight down, it does not swing or lift off, and no part of it rests on the bench. If the loop sits on the tip of the hook or the coat rests on the bench, close on the loop again, lift it just off the hook, and put it on again.

Expected state: one coat hangs on coat hook one. One coat lies at the coat spot. The other two hooks are empty. Both grippers are back and clear.

Step 2: Hang the second coat

Goal: the second coat hangs on coat hook two, and the coat spot is empty.

2.1 Lift the second coat by its loop

Close a gripper on the loop of the coat left at the coat spot.

Lift the coat straight up off the bench, and no higher.

Pull it forward into the front lane. Do not swing it. Do not drag it along the bench.

Check: the coat hangs straight down from its loop and the coat spot is empty. Nothing came up with it. If the coat hangs by a sleeve or a hem instead of the loop, lay it back down flat, close on the loop, and lift it again.

2.2 Carry it to the hook

Carry the coat along the front lane to coat hook two in a low carry. Keep it hanging straight the whole way.

Do not carry it across the back of the bench and do not carry it over the bowl.

Bring it in front of the hook. Then lift the loop straight up, level with the hook.

Check: the coat hangs straight and still in front of coat hook two, clear of the first coat. If it brushes the first coat, move it a little away and hold it still before you go on.

2.3 Put the loop on the hook

Move the loop straight back over the hook, so the hook goes through it.

Lower the loop until it sits all the way down on the hook and the coat takes its own weight.

Open the gripper and pull it back away from the coat.

Check: the second coat is hanging right and hanging clear, beside the first coat and not on top of it. The first coat has not been knocked off its hook. If the two coats hang on one hook, or one has come off, close on the loop again, lift it just off, and put it on the right hook.

Expected state: both coats hang on the two coat hooks, side by side and clear of each other. The coat spot is empty. The bag hook is empty. Both grippers are back and clear.

Step 3: Put the first shoe on the rack

Goal: one shoe stands upright on the rack shelf, and the shoe spot holds one shoe.

3.1 Lift the first shoe

Close a gripper on one shoe at the shoe spot, around the middle of the shoe.

Lift it straight up off the bench, and no higher.

Pull it forward into the front lane. Keep it upright, sole down.

Check: the shoe hangs upright and level in the gripper, sole down. The other shoe has not moved. If the shoe hangs upside down or by a lace, put it back down on the bench, close on the middle again, and lift it again.

3.2 Carry it to the rack

Carry the shoe along the front lane to the left end of the bench in a low carry. Keep it upright the whole way.

Do not carry it across the back of the bench and do not carry it over the bowl.

Then lower it down to the rack shelf, clear of the bench and clear of the mat.

Check: the shoe is over the rack shelf, still upright, sole down. No part of the arm rests on the rack or on the bench. If the rack moves, stop and report it.

3.3 Stand it on the shelf

Lower the shoe straight down onto the rack shelf.

Stand it flat on its sole, fully on the shelf, with its toe facing out to the front.

Open the gripper and pull it back away from the shoe.

Check: the shoe is standing right on the rack. It stands flat on its sole, fully on the shelf, toe to the front, and it does not fall or rock when the gripper lets go. If it tips or hangs over the front edge, close on it again, lift it just off the shelf, and stand it down again.

Expected state: one shoe stands on the rack shelf. One shoe lies at the shoe spot. Both coats still hang. Both grippers are back and clear.

Step 4: Pair the second shoe beside it

Goal: both shoes stand paired on the rack shelf, and the shoe spot is empty.

4.1 Lift the second shoe

Close a gripper on the shoe left at the shoe spot, around the middle of the shoe.

Lift it straight up off the bench, and no higher.

Pull it forward into the front lane. Keep it upright, sole down.

Check: the shoe hangs upright and level in the gripper and the shoe spot is empty. If it hangs by a lace, put it back down, close on the middle, and lift it again.

4.2 Carry it to the rack

Carry the shoe along the front lane to the left end of the bench in a low carry. Keep it upright the whole way.

Do not carry it across the back of the bench and do not carry it over the bowl.

Then lower it down to the rack shelf, beside the shoe that is already there.

Check: the shoe is over the free part of the shelf, still upright, and it is not over the first shoe. If it hangs over the first shoe, move it sideways before you lower it.

4.3 Stand it beside the first

Lower the shoe straight down onto the rack shelf, next to the first shoe.

Stand it flat on its sole, facing the same way as the first shoe, with the two heels level and a finger's width between them.

Open the gripper and pull it back away from the shoe.

Check: the two shoes are paired. They stand side by side, both toes to the front, heels level, with a small gap between them, and neither one rocks when the gripper lets go. The first shoe has not been pushed off the shelf. If they are crossed, one behind the other, or touching hard, close on the second shoe again, lift it just off the shelf, and stand it down again.

Expected state: both shoes stand paired on the rack shelf. The shoe spot is empty. The coat spot is empty. Both grippers are back and clear.

Step 5: Put the keys in the bowl

Goal: the whole key set lies in the bowl, and the key spot is empty.

5.1 Lift the keys

Close a gripper on the key ring at the key spot.

Lift the whole set straight up off the bench, and no higher.

Pull it forward into the front lane. Do not swing it. Do not drag it along the bench.

Check: the whole set hangs from the ring in the gripper and the key spot is empty. No key is left on the bench. If a key slid off the ring, stop and report it.

5.2 Carry them to the bowl

Carry the keys along the front lane to the right end of the bench in a low carry. Keep them low and still.

Then move them straight back to the bowl. Do not swing them over the bowl.

Hold them over the middle of the bowl, just above its rim.

Check: the keys hang still over the middle of the bowl. They are not over the bench top and not over the bag. If they swing, hold them still and let them settle before you lower them.

5.3 Lower them in

Lower the keys straight down into the bowl until they rest on its floor.

Open the gripper and pull it back away from the bowl.

Do not touch the rim of the bowl and do not push the bowl.

Check: the keys are in the bowl. The whole set lies inside, below the rim, with no key hanging over the edge and none on the bench. The bowl has not moved. If a key hangs over the rim, close on the ring again, lift the set just off, and lower it in again.

Expected state: the keys lie in the bowl. The key spot is empty. Both coats hang and both shoes are paired. Both grippers are back and clear.

Step 6: Hang the bag

Goal: the bag hangs on the bag hook, and the bench top is clear.

6.1 Lift the bag by its strap

Close a gripper on the strap of the bag at the bag spot, at the middle of the strap.

Lift the bag straight up off the bench, and no higher.

Pull it forward into the front lane. Keep it hanging straight from the strap.

Check: the bag hangs straight down from its strap and the bag spot is empty. The strap is not caught under the bag. If the bag hangs at an angle or by its body, put it back down flat, lay the strap clear, close on the middle of the strap, and lift it again.

6.2 Carry it to the hook

Carry the bag along the front lane to the bag hook in a low carry. Keep it hanging straight the whole way.

Do not carry it across the back of the bench and do not carry it over the bowl.

Bring it in front of the hook. Then lift the strap straight up, level with the hook.

Check: the bag hangs straight and still in front of the bag hook, clear of the coats and clear of the bowl. If it swings, hold it still and let it settle before you go on.

6.3 Put the strap on the hook

Move the strap straight back over the hook, so the hook goes through it.

Lower the strap until it sits all the way down on the hook and the bag takes its own weight.

Open the gripper and pull it back away from the bag.

Check: the bag is hanging right and hanging clear. The strap sits all the way over the hook, the bag hangs straight down, it does not swing or lift off, and it does not rest on the bench top or on the bowl. If the strap sits on the tip of the hook or the bag rests on the bench, close on the strap again, lift it just off the hook, and put it on again.

Expected state: two coats and the bag hang on the three hooks. The keys lie in the bowl. Both shoes stand paired on the rack. The bench top is clear. Both grippers are back and clear.

Step 7: Square the mat

Goal: the mat lies flat on the floor in front of the bench, with its near edge straight and parallel to the bench front edge.

Push the mat only. You do not lift it, fold it, or straighten anything else in this step.

7.1 Look at the mat

Look at the mat from the front, with the front edge of the bench in the same view.

See which end of the near edge is further from the bench, and by how much.

Pick the corner you will push.

Check: the whole mat is in view, it lies flat, and no corner is folded under. If a corner is folded under, stop and report it. You do not lift the mat to fix it.

7.2 Push it square

Bring a closed gripper down to the mat's edge, close to the floor, at the corner you picked.

Push the mat flat along the floor in short pushes, until its near edge runs straight across, parallel to the front edge of the bench.

Push the other corner the same way if the mat still runs off square.

Check: the near edge of the mat is square to the bench, and both ends of that edge are the same distance from the bench. The mat still lies flat, with no bunching and no corner folded under. If it has bunched up, push it flat again from the same edge.

7.3 Set the gap

Look at the strip of floor between the bench front edge and the mat's near edge.

Push the mat forward or back in short pushes, until that strip is the same width all the way across.

Pull the gripper back away from the mat and up off the floor.

Check: the mat is square, it lies flat, and the gap to the bench is even end to end. No part of an arm rests on the mat or on the floor. If the mat turned off square while you set the gap, push a corner again until it runs straight.

Expected state: the mat lies flat and square in front of the bench. Nothing else has moved. Both grippers are back and clear.

Step 8: End the episode

Check the coats: one hangs on each coat hook, straight down, side by side, clear of each other and clear of the bench.

Check the shoes: both stand on the rack shelf, upright, facing the same way, heels level, paired, and fully on the shelf.

Check the keys: the whole set lies in the bowl, below the rim, with no key outside it.

Check the bag: it hangs on the bag hook, straight down, clear of the coats, the bowl, and the bench top.

Check the bench: the top is clear, with nothing left at the coat spot, the shoe spot, the key spot, the bag spot, or the front lane.

Check the mat: it lies flat and square, with an even gap to the bench.

Check the fixtures: the bench, the hook board, the bowl, and the shoe rack stand where they started, and nothing is on the floor but the shoes on the rack and the mat.

Check the base has not moved.

Fix any failed check before you end.

Send both arms home with the grippers open. Then stop recording.

After the episode: reset the workspace

This reset is not recorded.

1. Take both coats off their hooks. Lay them flat at the coat spot, one on top of the other, each one with its loop up and clear.
2. Take the bag off its hook. Lay it at the bag spot, past the keys, with its strap laid out clear and not under the bag.
3. Take the keys out of the bowl. Lay the whole set at the key spot.
4. Take both shoes off the rack. Lay them loose at the shoe spot, apart from each other and not paired.
5. Push the mat off square again, so it starts crooked, and check it lies fully flat with no corner folded under.
6. Wipe the bench top and dry it. Wipe the bowl inside and the rack shelf, and dry them.
7. Check all three hooks are empty, tight on the board, and not bent.
8. Check the bowl is empty and stands at the back right of the bench top.
9. Check the shoe rack is empty, standing square on the floor at the left end of the bench, and clear of the bench.
10. Change a coat if its loop is torn, worn loose, or will not stand up when the coat lies flat.
11. Change the bag if its strap is torn or will not stay over the hook in a correct hang.
12. Change a shoe if it will not stand upright on its own sole on a flat shelf.
13. Change the mat if it will not slide flat, if it bunches when it is pushed, or if a corner will not lie down.
14. Clear anything that ended up on the floor and dry the floor before the next episode.

15. Check the base is still locked and parked square. Then do the Base positioning steps and both Setup checklists again.

SOP violations

These are actions that break the SOP. They are reviewed side by side in the review tool.

How to record a violation in review

For each violation, write down the start timestamp, the violation name, and the SOP rule broken. The visible cue is what the reviewer sees. The coaching note is for retraining. It is not an annotation label.

Episode handling

Tag every violation with its timestamp and its name. Keep the episode with the violation tag. Do not delete it just because a rule was broken.

Violations

Violation: Base moved during the episode

Visible cue: the bench moves in the frame, the bench edge changes angle in the frame, or the base rolls, slides, or turns at any time after recording starts.

SOP rule broken: Steps 1 to 8. The base is parked and locked before recording and stays still for the whole episode.

Coaching note: park it, lock it, push it to test it, then start recording. A base that moves is a lost episode.

Violation: Lifted high or hit the hook board

Visible cue: a coat, a shoe, the keys, or the bag is lifted well above the bench instead of just off what it lies on. Or a wrist or forearm knocks, scrapes, or rests on the hook board. The only lifts that may go up are the loop and the strap going onto their hooks.

SOP rule broken: Steps 1.1, 1.2, 2.1, 2.2, 3.1, 3.2, 4.1, 4.2, 5.1, 5.2, 6.1, and 6.2. Lift just off, come forward into the front lane, and carry low and level.

Coaching note: up a little, forward, then along. Only the loop and the strap go up, and only at the hook.

Violation: Carried outside the front lane

Visible cue: a carry goes across the back of the bench or over the bowl instead of along the front lane. Or a thing is put down in the lane in the middle of a carry.

SOP rule broken: Steps 1.2, 2.2, 3.2, 4.2, 5.2, and 6.2. Every carry goes along the front lane, and nothing is ever put down in it.

Coaching note: the lane is the road. Nothing crosses the back of the bench, and nothing parks in the lane.

Violation: Leaned on or moved a fixture

Visible cue: a gripper, wrist, or forearm rests on the bench, the hook board, a hook, the bowl, or the shoe rack. Or the bowl slides, or the rack tips or slides, or the hook board moves.

SOP rule broken: Steps 1 to 7. Nobody leans on, pushes, drags, or lifts the bench, the hook board, a hook, the bowl, or the shoe rack.

Coaching note: the arm holds itself up. If something moves when you touch it, you touched it too hard.

Violation: Held by the wrong part

Visible cue: a gripper closes on a coat's sleeve, hem, collar, or body instead of its loop. Or on the bag's body instead of its strap. Or on a shoe's lace or tongue instead of the middle of the shoe. Or on a single key instead of the ring.

SOP rule broken: Steps 1.1, 2.1, 3.1, 4.1, 5.1, and 6.1. Coats by the loop, the bag by the strap, a shoe by its middle, the keys by the ring.

Coaching note: one hold for each thing, every time. The right hold is the one that hangs straight.

Violation: Dragged instead of lifted

Visible cue: a coat, a shoe, the keys, or the bag slides along the bench top instead of coming straight up off it before it moves.

SOP rule broken: Steps 1.1, 2.1, 3.1, 4.1, 5.1, and 6.1. Every pick lifts the thing straight up off the surface first.

Coaching note: up first, then along. A dragged thing pulls the next one with it.

Violation: Two things moved at once

Visible cue: both coats come off the stack together, both shoes are held at the same time, or one thing is still held while another is picked up.

SOP rule broken: Steps 1 to 7. Only one thing is held at a time, and it is let go before the next one starts.

Coaching note: one thing, all the way to its place, then let go. Then the next one.

Violation: Not hanging right

Visible cue: a loop or the strap sits on the tip of a hook, hangs half over it, or comes off when the gripper lets go. Or a coat or the bag is left on the bench instead of on its hook.

SOP rule broken: Steps 1.3, 2.3, and 6.3. The loop or the strap sits all the way over the hook and takes its own weight before you let go.

Coaching note: all the way over, then watch it take its own weight before you open the gripper.

Violation: Not hanging clear

Visible cue: a hung coat or the bag rests on the bench top, on the bowl, or on the wall, a sleeve or a strap lies across the bench, or the two coats hang tangled or both on one hook.

SOP rule broken: Steps 1.3, 2.3, and 6.3. Each thing hangs free of the bench, the wall, and the thing next to it.

Coaching note: three hooks, three things, nothing touching. If it leans on something, it is not hung.

Violation: Shoes not paired

Visible cue: at the end of Step 4 the two shoes stand crossed, one behind the other, one on top of the other, facing different ways, with heels well out of line, or one shoe is left off the rack.

SOP rule broken: Steps 3.3 and 4.3. Both shoes stand side by side, facing the same way, heels level, with a finger's width between them.

Coaching note: side by side, same way, heels in line. A pair is read in one look.

Violation: Shoe not standing right

Visible cue: a shoe lies on its side, stands on the front edge of the shelf with part hanging over, rocks when the gripper lets go, or falls off the rack.

SOP rule broken: Steps 3.3 and 4.3. Stand it flat on its sole, fully on the shelf, and watch it settle before you let go.

Coaching note: sole down, fully on, then let go. A shoe that rocks falls after you leave.

Violation: Keys not in the bowl

Visible cue: the keys are dropped from above the bowl, land on the bench, hang over the rim, bounce out, or a key ends up outside the bowl.

SOP rule broken: Step 5.3. Lower the whole set straight down into the bowl until it rests on the floor of the bowl.

Coaching note: lower them in, do not drop them in. Keys bounce.

Violation: Mat lifted or bunched

Visible cue: the gripper lifts the mat off the floor, digs under it, folds it, or pushes it so hard that it bunches up or rolls.

SOP rule broken: Steps 7.2 and 7.3. The mat is pushed flat along the floor and is never lifted.

Coaching note: short pushes at the edge, close to the floor. The mat stays down.

Violation: Mat left crooked

Visible cue: at the end of Step 7 the near edge of the mat does not run parallel to the bench front edge, or the gap to the bench is clearly wider at one end than the other.

SOP rule broken: Steps 7.2 and 7.3. The near edge runs straight across, and the gap is even end to end.

Coaching note: look along the edge, not at the middle. If the gap changes, the mat is not square.

Violation: Done in the wrong order

Visible cue: a shoe moves before both coats hang, the keys move before both shoes are paired, the bag moves before the keys are in the bowl, or the mat is pushed before the bench top is clear.

SOP rule broken: Steps 1 to 7. Coats, then shoes, then keys, then bag, then the mat.

Coaching note: one thing to its place before the next one starts. The mat is always last.

Violation: Bench left with something on it

Visible cue: at the end of the episode a coat, a shoe, the keys, or the bag is still on the bench top, or something is left standing in the front lane.

SOP rule broken: Step 8. The bench top is clear at the end, with nothing left at any spot and nothing in the lane.

Coaching note: the bench is empty when you are done. If it is on the bench, it is not put away.

Violation: Something dropped or knocked over

Visible cue: a coat, a shoe, the keys, or the bag slips out of a gripper, falls on the bench or on the floor, or is knocked over by an arm going past. Or the bowl is tipped.

SOP rule broken: Steps 1 to 7. One thing moves at a time, and each one is put down or let go under control.

Coaching note: close on the hold all the way before you lift, and put it down before you let go.

Violation: Failed check not retried

Visible cue: a check in a step clearly fails, and the episode carries on with no retry. For example, a loop sits on the tip of a hook, a shoe rocks on the shelf, a key hangs over the bowl rim, or the mat still runs crooked.

SOP rule broken: Steps 1 to 7. Do the retry written under each failed check before you move on.

Coaching note: a check is only worth doing if you do the retry. Fix it where it happened.

Violation: Wrong ending

Visible cue: the episode ends with a coat, a shoe, the keys, or the bag out of place, the bench top not clear, or the mat crooked. Or an arm is not home, or a gripper is not fully open.

SOP rule broken: Step 8. Check the coats, the shoes, the keys, the bag, the bench, the mat, the fixtures, and the base. Then send both arms home with the grippers open and stop recording.

Coaching note: check first. Going home is the last thing the arms do.

Failures that are not violations

These failures are not caused by how the task was run. Log them as system issues, throw the episode away, and never use them for coaching.

Recording stopped or paused during the episode (recording system).

Camera dropped frames or lost its feed (capture system).

Camera cannot see a hook, cannot see inside the bowl, cannot see the rack shelf, or cannot see the near edge of the mat, so nobody can tell if a thing was placed right.

Hardware fault on an arm: gripper failure, drift, a crash caused by the controller, or a motor error.

Base fault: a lock that will not hold, or a base that rolls, slides, or turns with the lock on.

Faulty hook board: one that moves, turns, sags, or comes loose under a correctly placed loop or strap.

Faulty hook: one bent, too short, or too smooth to hold a loop or a strap in a correct hang.

Faulty coat: one with a torn or missing loop, a loop that will not stand up when the coat lies flat, or a coat too heavy for its hook.

Faulty bag: one with a torn strap, a strap too short or too wide for the hook, or a bag that spills its contents when it hangs.

Faulty shoe: one that will not stand upright on its own sole on a flat shelf, or one wet or dirty underneath.

Faulty shoe rack: one that tips, slides, or sags under a correctly placed shoe, or a shelf too narrow for the two shoes side by side.

Faulty bowl: one that slides under a correct lowering, that is too small or too shallow for the whole key set, or that is cracked.

Faulty keys: a ring that comes open, or a key that is loose off the ring at the start.

Faulty mat: one that will not slide flat on the floor, that bunches or rolls when it is pushed at its edge, or that has a corner that will not lie down.

Faulty floor: one that is wet, sticky, or uneven, so the mat will not slide or will not lie flat.

A place turns out to be too far for the arm with the base parked correctly, so a coat, a shoe, the keys, the bag, the bowl, a hook, the rack shelf, or the mat cannot be reached without stretching or folding the arm.

Annotation subtasks (from SOP)

1. Lift the first coat off the bench by its loop
2. Carry the first coat along the front lane to the hook
3. Put the first coat's loop on the hook
4. Lift the second coat off the bench by its loop
5. Carry the second coat along the front lane to the hook
6. Put the second coat's loop on the hook
7. Lift the first shoe off the bench
8. Carry the first shoe along the front lane to the rack
9. Stand the first shoe on the rack shelf
10. Lift the second shoe off the bench
11. Carry the second shoe along the front lane to the rack
12. Stand the second shoe beside the first and pair them
13. Lift the keys off the bench by the ring
14. Lower the keys into the bowl
15. Lift the bag off the bench by its strap
16. Carry the bag along the front lane to the hook
17. Put the bag's strap on the hook
18. Push the mat square to the bench
19. Set the gap between the mat and the bench
20. Send both arms home and end the episode